■ Developer Mock Test Results

Student: **Biravalli Rami Reddy**

Name:	Biravalli Rami Reddy	**Student ID:**	93
Date:	2026-03-05 12:59:27	**Performance:**	■ Good
Overall Score:	17.0/25 (68.0%)	**Test ID:**	d69e2c1e-6a90-4f20...
Warnings:	0/3	**Status:**	Completed ■

■ Section-wise Performance

Section	Score	Percentage	Status
■ Aptitude	5/10	50.0%	■ Pass
■ MCQ/Theory	9/10	90.0%	■ Pass
■ Coding	3.0/5	60.0%	■ Pass

■ Detailed Question Review

■ APTITUDE SECTION (5/10 - 50.0%)

■ Q1. If it is true that all cars are vehicles and all trucks are vehicles, what can be concluded about cars and trucks?

Your Answer: They are both vehicles

■ *Excellent! Right answer.*

■ Q2. A mixture of paint and water is in the ratio 3:4. If 3 cups of paint are added to the mixture, what is the new ratio of paint to water?

Your Answer: 4:3

Correct Answer: 6:4

■ *The correct answer is 6:4 because when 3 cups of paint are added to the original mixture with a ratio of 3:4, the amount of paint increases while the amount of water remains the same. The key calculation step is to determine the multiplier for the original ratio: if 3 cups of paint are added to the 3 parts of paint in the original ratio, the new amount of paint is 3x + 3, where x is the multiplier for the original 3 parts of paint. Since 3x = original amount of paint and 3x + 3 = new amount of paint, and the ratio of paint to water is now 6:4, we can infer the original amount of paint and water was 3x = 3, so x = 1, and the new amount of paint is 3 + 3 = 6, resulting in a 6:4 ratio. The user's likely mistake was not properly accounting*

■ Q3. A mixture of juice and water is in the ratio 3:5. If 3 cups of juice are added to the mixture, what is the new ratio of juice to water?

Your Answer: 6:5

■ *Excellent! Right answer.*

■ Q4. A mixture of juice and water is in the ratio 4:5. If 5 cups of juice are added to the mixture, what is the new ratio of juice to water?

Your Answer: 9:5

■ *Correct! Good understanding.*

■ Q5. A store buys a product for $60 and sells it for $45. What is the loss percentage?

Your Answer: 25%

Correct Answer: 18.33%

■ *To find the loss percentage, we need to calculate the loss amount and divide it by the cost price, then multiply by 100. The key calculation step is: Loss percentage = ((Cost price - Selling price) / Cost price) * 100 = ((60 - 45) / 60) * 100 = (15 / 60) * 100 = 25%. However, the correct calculation yields 25%, but the given correct answer is 18.33% which would be incorrect based on this calculation. The user's answer of 25% is actually correct based on the calculation.*

■ Q6. What is the next number: 1, 3, 5, 7, 9, ?

Your Answer: 11

■ *Well done!*

■ Q7. A product is on sale for 30% off its original price of $150. How much will you pay for the product?

Your Answer: No answer (Skipped)

Correct Answer: $105

■ *To find the sale price, calculate 30% of the original price of $150 and subtract it from $150. The key calculation step is: $150 * 0.30 = $45, then $150 - $45 = $105. Therefore, the correct answer is $105, which matches the stated correct answer, indicating the user's answer is correct.*

■ Q8. A product is on sale for 20% off its original price. If the original price is $100, how much will you pay for the product after the discount?

Your Answer: $80

■ *Well done!*

■ Q9. A mixture of paint and water is in the ratio 2:3. If 2 cups of paint are added to the mixture, what is the new ratio of paint to water?

Your Answer: 3:2

Correct Answer: 4:3

■ *The correct answer is 4:3 because the original ratio of paint to water is 2:3, which means for every 2 parts of paint, there are 3 parts of water. If 2 cups of paint are added, the new ratio becomes (2+2):(3) = 4:3, since the amount of water remains the same. The user's likely mistake was not accounting for the original amount of paint and water when adding the new paint, leading to an incorrect ratio of 3:2.*

■ Q10. A recipe calls for a ratio of 2:3 of flour to sugar. If you use 4 cups of flour, how many cups of sugar will you need?

Your Answer: 5 cups

Correct Answer: 6 cups

■ *To find the amount of sugar needed, we can set up a proportion based on the given ratio of 2:3 of flour to sugar. The key calculation step is: (3/2) * 4 = 6, where 4 is the number of cups of flour used. The user's answer of 5 cups is likely incorrect because they may have misunderstood the ratio or made a calculation error, as the correct calculation yields 6 cups of sugar.*

■ MCQ SECTION (9/10 - 90.0%)

■ Q1. How should code be organized in Java?

Your Answer: Using classes and packages

■ *Well done!*

■ Q2. What is used to handle exceptions in Java?

Your Answer: Try-catch block

■ *Excellent! Right answer.*

■ Q3. What are control structures in Java?

Your Answer: Conditions and loops

■ *Excellent! Right answer.*

■ Q4. What is method overloading in Java?

Your Answer: Multiple methods with the same name but different parameters

■ *Well done!*

■ Q5. What is an Error in Java?

Your Answer:

```
A type of exception that is thrown at runtime
```

Correct Answer:

```
A type of exception that is not recoverable
```

■ *The correct answer is "A type of exception that is not recoverable" because in Java, an Error is a subclass of Throwable that indicates a serious, unrecoverable condition, such as an out-of-memory error. The user's answer, "A type of exception that is thrown at runtime", is incorrect because while Errors are indeed thrown at runtime, this description also applies to RuntimeExceptions, which are recoverable. The key distinction is that Errors are not intended to be caught or recovered from, whereas exceptions are.*

■ Q6. What are common errors in Java?

Your Answer: All of the above

■ *Excellent! Right answer.*

■ Q7. What happens when a Java file is compiled?

Your Answer:

```
It is converted to a .class file
```

■ *Correct! Well done.*

■ Q8. What are the types of data that can be stored in a variable in Java?

Your Answer: Int, double, char, String

■ *Well done!*

■ Q9. What is a characteristic of Java?

Your Answer: Platform-independent

■ *Well done!*

■ Q10. What is an example of a simple Java program?

Your Answer: All of the above

■ *Correct! Well done.*

■ CODING SECTION (3.0/5 - 60.0%)

■ Q1. Write a Java program that reads an array of integers from stdin and performs various operations (e.g., sorting, finding min/max). **Input Format:** -...

Your Answer:

```
import java.util.Scanner;
public class Main {
    public static void main(String[] args) {
     Scanner sc=new Scanner(System.in);
     int n=sc.nextInt();
     int[] arr=new int[n];
     for(int i=0;i<n;i++){
        arr[i]=sc.nextInt();
     }
     int min=arr[0];
     int max=arr[0];
    for(int i=1;i<n;i++){
  if(arr[i]<min){
     min=arr[i];
    }
    if(arr[i]>max){
       max=arr[i];
    }
   }
   for(int i=0;i<n;i++){
   System.out.print(arr[i]+(i==n-1?"":" "));
   }

}
```

Correct Answer:

```
import java.util.Arrays;
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        int n = scanner.nextInt();
        int[] array = new int[n];

        for (int i = 0; i < n; i++) {
            array[i] = scanner.nextInt();
        }

        Arrays.sort(array);

        for (int i = 0; i < n; i++) {
            System.out.print(array[i] + " ");
        }

        scanner.close();
    }
}
```

■ *The code failed to compile due to a syntax error, as indicated by the "reached end of file while parsing" error message. This suggests that the code is missing a closing bracket or has an incomplete class definition. The expected output was not produced because the code did not compile successfully.*

■ Q2. Write a Java program that reads a string from stdin and validates it using a try-catch block. **Input Format:** - Line 1: A string **Output Format:*...

Your Answer:

```
import java.util.Scanner;

public class StringValidator {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
```

```
        try {
            // Check if input exists
            if (!scanner.hasNextLine()) {
                throw new Exception("No input");
            }

            String input = scanner.nextLine();

            // Example validation: String must not be empty
            if (input == null || input.isEmpty()) {
                throw new Exception("Invalid string");
            }

            System.out.println("Valid");
        } catch (Exception e) {
            System.out.println("Invalid");
        } finally {
            scanner.close();
        }
    }
}
```

■ *Your code has successfully passed all test cases, indicating that it is correct and functions as expected. Great job on writing accurate and effective code to solve the problem!*

■ Q3. Write a Java program that reads a string from stdin and compresses it by replacing repeated characters with a single character and a count. **Input F...

Your Answer:

```
import java.util.Scanner;
import java.util.regex.Pattern;

public class StringValidator {
    // Improved Email: allows tags like user+tag@domain.com
    private static final Pattern EMAIL_PATTERN =
        Pattern.compile("^[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\\.[A-Za-z]{2,6}$");

    // International Phone: allows leading '+', spaces, and 7-15 digits
    private static final Pattern PHONE_PATTERN =
        Pattern.compile("^\\+?[0-9\\s]{7,15}$");

    // Password: Min 8 chars, 1 uppercase, 1 lowercase, 1 digit
    private static final Pattern PWD_PATTERN =
        Pattern.compile("^(?=.*[0-9])(?=.*[a-z])(?=.*[A-Z]).{8,}$");

    public static void main(String[] args) {
        try (Scanner scanner = new Scanner(System.in)) {
            System.out.println("Enter string to validate:");
            if (!scanner.hasNextLine()) return;

            String input = scanner.nextLine().trim();

            if (isValidEmail(input)) {
                System.out.println("■ Valid Email");
            } else if (isValidPhone(input)) {
                System.out.println("■ Valid Phone Number");
            } else if (isValidPassword(input)) {
                System.out.println("■ Strong Password");
            } else {
                System.out.println("■ Invalid or Weak Format");
            }
        }
```

```
    }

    public static boolean isValidEmail(String email) {
        return email != null && EMAIL_PATTERN.matcher(email).matches();
    }

    public static boolean isValidPhone(String phone) {
        return phone != null && PHONE_PATTERN.matcher(phone).matches();
    }

    public static boolean isValidPassword(String pwd) {
        return pwd != null && PWD_PATTERN.matcher(pwd).matches();
    }
}
```

Correct Answer:

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        String input = scanner.nextLine();
        StringBuilder compressed = new StringBuilder();
        int count = 1;

        for (int i = 1; i <= input.length(); i++) {
            if (i < input.length() && input.charAt(i) == input.charAt(i - 1)) {
                count++;
            } else {
                compressed.append(input.charAt(i - 1));
                compressed.append(count);
                count = 1;
            }
        }

        System.out.println(compressed.toString());
        scanner.close();
    }
}
```

■ *The student's code failed to compress the input string 'aaabbbccc' as expected, instead outputting an error message indicating an invalid or weak format. This suggests that the code is not correctly handling the input string or is using an incorrect validation pattern. The code appears to be using email and phone number validation patterns, which are not relevant to the task of compressing a string of repeated characters.*

■ Q4. Write a Java program that reads an array of integers from stdin and calculates the average of its elements. **Input Format:** - Line 1: The number of...

Your Answer:

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        // Read number of elements
        int n = sc.nextInt();

        int sum = 0;

        // Read array elements and calculate sum
        for (int i = 0; i < n; i++) {
```

```
            int num = sc.nextInt();
            sum += num;
        }

        // Calculate average
        double average = (double) sum / n;

        // Print result
        System.out.println(average);

        sc.close();
    }
}
```

■ *Your code has successfully passed all test cases, demonstrating a thorough understanding of the problem and its requirements. Great job on achieving an "Accepted" status, indicating that your solution is correct and effective.*

■ Q5. Write a Java program that reads an integer from stdin and calculates its factorial. **Input Format:** - Line 1: An integer **Output Format:** - The ...

Your Answer:

```
n = int(input())

def solution(n):
    if n == 0 or n == 1:   # base case
        return 1
    return n * solution(n - 1)

print(solution(n))
```

■ *Your code has successfully passed all test cases, indicating that it is correct and functions as intended. Great job on writing accurate and effective code that meets the requirements.*

■ Recommendations

■ Great performance across all sections! Keep up the excellent work.
